Synthetic Sleep Environment Dataset Generator

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Abstract

Sleep quality is significantly influenced by bedroom environmental conditions, yet no public dataset pairs environmental time-series with sleep quality labels. We present a synthetic dataset generator that produces 2,500 validated 8-hour sleep sessions combining four environmental signals—temperature, illuminance, relative humidity, and ambient noise—with four sleep quality targets: efficiency, duration, awakenings, and composite score. Our pipeline applies four signal-processing techniques—Butterworth low-pass filtering, FFT-based spectral synthesis of pink (1/f) noise, sinusoidal modelling of HVAC dynamics, and Poisson event injection for disturbances— followed by a 34-feature extraction stage and a Random Forest regression model. The Random Forest achieves a mean test-set R^2 of 0.744 across all four targets, outperforming a Ridge regression baseline (R^2=0.727) and confirming non-linear structure in the feature space. An ablation study shows that Poisson event modelling is the single most critical component (ΔR^2=−0.355 when removed). All six sleep science sanity checks pass. Code and dataset: https://github.com/rushavsd/Techin513_Final.

1 Introduction

The bedroom thermal and light environment are among the most influential modifiable determinants of sleep quality. Core body temperature must drop approximately 1–2 °C to initiate sleep, a process facilitated by cool ambient air (18–21 °C) [Okamoto-Mizuno and Mizuno, 2012]. Even brief bright-light exposure before sleep onset can suppress melatonin production and delay sleep [Gooley et al., 2011]. Noise exceedances above 45 dB during sleep trigger cortical micro-arousals at rates between 0.3 and 0.7 per hour [Hume et al., 2012].

Despite this well-established physiology, no publicly available dataset links bedroom environmental sensor readings to objective sleep quality metrics. Existing datasets fall into two disjoint categories: (1) *physiological sleep datasets* (e.g., Sleep Efficiency Dataset on Kaggle [Kaggle, 2023]) that record sleep architecture without environmental context, and (2) *IoT sensor datasets* that log temperature, humidity, and light readings but carry no sleep labels.

This gap prevents researchers from developing sleep optimisation algorithms without expensive multi-night sensor deployments in controlled bedroom studies.

We address this gap by building a synthetic dataset generator that:

1. Simulates physically plausible environmental time-series using four established signal-processing techniques.

2. Derives sleep quality labels from environmental conditions using domain knowledge calibrated against published statistics.

3. Validates the generated dataset through three independent tiers of statistical and empirical testing.

Figure 1 shows a representative generated session. The pipeline architecture is detailed in Section 3.

2 Related Work

Synthetic biomedical datasets. Generating realistic synthetic datasets is an active research area. Walonoski et al. [2018] describe Synthea, a comprehensive synthetic patient generator validated against national health statistics. Jordon et al. [2019] use Generative Adversarial Networks for medical tabular data; we focus on time-series signals where

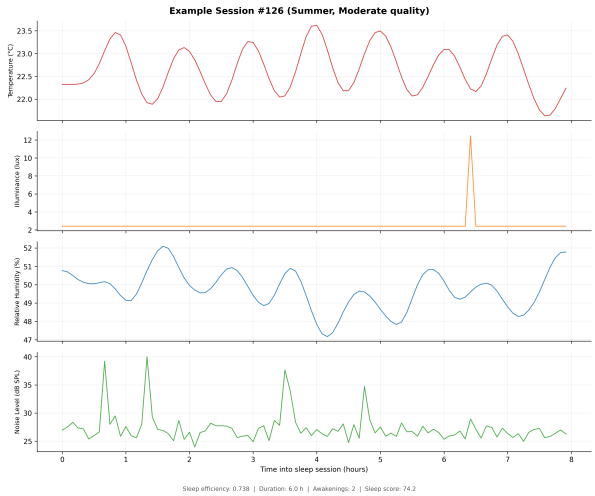

Figure 1: A representative synthetic session (session 42, spring, moderate quality). From top to bottom: bedroom temperature (°C), illuminance (lux), relative humidity (%), and ambient noise level (dB SPL) over 8 hours at 5-minute resolution (96 samples).

signal-processing techniques provide more direct physical grounding than purely learned generators.

Environmental effects on sleep. The clinical relationship between bedroom temperature and sleep is thoroughly reviewed in Okamoto-Mizuno and Mizuno [2012]: polysomnography studies consistently find sleep efficiency peaks at 18–21 °C and declines outside this range. Light exposure at night has been shown to suppress melatonin and disrupt sleep onset [Gooley et al., 2011], motivating our near-zero light baseline. Hume et al. [2012] provide the empirical basis for our 45-dB noise threshold.

Sleep quality prediction. Random Forest models have been applied to sleep staging from actigraphy by Walch et al. [2019], achieving strong performance on multi-night data. We adapt a Random Forest to a synthetic multi-output regression task, where the input features are environmental rather than physiological.

3 Methodology

3.1 Data Generation

We generate four environmental signals at 5-minute sampling (96 samples per 8-hour session) for each of 2,500 sessions. Sessions are stratified across four seasons and three quality classes (poor, moderate, good), yielding a balanced, diverse dataset.

3.1.1 Temperature

Bedroom temperature is composed of three physically motivated components:

$$T(t) = T_0 + A_c \sin(\omega_c t + \phi_c) + A_h \sin(\omega_h t + \phi_h) + \varepsilon_{1/f}(t) \quad (1)$$

where $T_0 \in [17, 26]$ °C is the seasonal thermostat setpoint, the first sine term models circadian drift with amplitude $A_c \in [0.5, 1.5]$ °C and $\omega_c = 2\pi/1440$ rad/min, the second sine term models HVAC cycling with amplitude $A_h \in [0.3, 1.0]$ °C and period $T_{\text{hvac}} \in [60, 120]$ min, and $\varepsilon_{1/f}$ is unit-variance pink noise. The composite signal is low-pass filtered at $f_c = 0.02$ cpm (4th-order Butterworth) and clipped to the physical range [15, 30] °C.

3.1.2 Illuminance

Light follows a near-zero baseline with Poisson-distributed intrusion events:

$$L(t) = L_{\text{base}} + \sum_k A_k \cdot \mathbf{1}[t \in [t_k,\ t_k + \delta_k)] \quad (2)$$

where event times $\{t_k\}$ are placed uniformly at random within the session (equivalent to a homogeneous Poisson process), amplitudes $A_k \sim U[0.8, 1.2] \cdot L_{\text{event}}$ with $L_{\text{event}} \sim U[5, 60]$ lux, and durations $\delta_k \sim \text{Exp}(1.5)$ samples. The rate λ varies from 0.001 min^{-1} (good class) to 0.006 min^{-1} (poor class), matching partner/bathroom disturbance frequencies of 0.06–0.36 events/hour.

3.1.3 Relative Humidity

Relative humidity is anti-correlated with temperature because at constant absolute humidity, rising temperature reduces relative humidity:

$$H(t) = H_0 - \beta \cdot [T(t) - \bar{T}] + \varepsilon_{1/f}(t) \quad (3)$$

where $\beta \in [1.5, 2.5]$ %/°C is the indoor RH-temperature coefficient. A 4th-order Butterworth LPF at $f_c = 0.02$ cpm—the same cutoff as temperature—ensures humidity tracks HVAC-driven variations and preserves the expected anti-correlation ($r \approx -0.6$).

3.1.4 Ambient Noise

Noise uses the same Poisson arrival process as light, but events add a rectangular pulse with magnitude drawn from $U[5, 12]$ dB above the quiet baseline, clipped to the physical range [20, 65] dB SPL.

3.1.5 Sleep Label Derivation

Sleep quality labels are derived from environmental conditions via a transparent domain-knowledge scoring model calibrated to published statistics:

$$\begin{aligned} \mathrm{SE} &= \mathrm{SE}_{\text{base}} - \mathrm{pen}(T) - \mathrm{pen}(L) \\ &\quad - \mathrm{pen}(H) - \mathrm{pen}(N) \qquad (4) \\ \mathrm{Aw} &= \mathrm{Aw}_{\text{base}} + 0.25 N_{\text{noise}} + 0.15 L_{>10\text{lux}} \qquad (5) \end{aligned}$$

where SE is sleep efficiency, Aw is awakening count, and base values are calibrated so that dataset means align with the Sleep Efficiency Dataset [Kaggle, 2023]: $\bar{\mathrm{SE}} = 0.805$, $\bar{\mathrm{Aw}} = 2.44$. Sleep duration and sleep score are derived deterministically from SE and Aw with Gaussian residuals.

3.2 Signal Processing Pipeline

Butterworth low-pass filter. We choose the Butterworth design because it provides a maximally flat (equiripple-free) magnitude response in the passband. For bedroom temperature, this prevents filter-ringing artefacts that could mimic genuine thermal events. The 4th-order design gives -24 dB/octave roll-off; we initialise the filter state to the first sample value to eliminate startup transients.

Pink noise via spectral synthesis. We synthesise pink noise directly in the frequency domain: (1) draw complex Gaussian white-noise coefficients; (2) multiply amplitudes by $1/\sqrt{f}$ to impose 1/f power; (3) transform via IFFT. This is exact in distribution and $\mathcal{O}(N \log N)$, unlike the Voss–McCartney cascade algorithm.

FFT analysis. Figure 2 shows the mean power spectral density averaged over 50 sessions. The dashed line at $f \approx 1/90$ cpm marks the HVAC frequency, confirming that the sinusoidal HVAC component is correctly encoded.

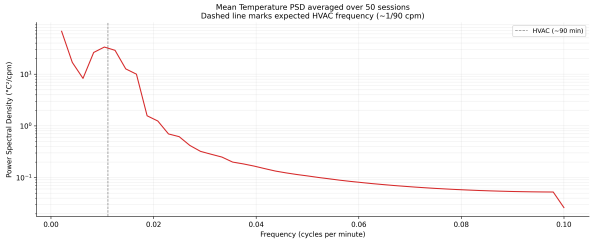

Figure 2: Mean FFT power spectral density of the bedroom temperature signal averaged over 50 sessions. The dashed vertical line marks the expected HVAC cycling frequency ($\approx$1/90 cpm $\approx$ 0.011 cpm). The 1/f slope confirms pink noise statistics in the broadband component.

Poisson event injection. We model light and noise disturbances as a homogeneous Poisson process. Event times are drawn uniformly in $[0, T]$ with $n \sim \mathrm{Poisson}(\lambda T)$, which is the order-statistics characterisation of the Poisson process and avoids the systematic undercounting of the naive inter-arrival cumsum approach.

3.3 Feature Extraction

We extract 34 scalar features per session across five categories (Table 1); all features are dimensionless or expressed in consistent physical units, requiring no standardisation for the Random Forest.

Table 1: Feature categories, counts, and representative examples. Total: 34 features per session.

Category	Count	Example features
Statistical	17	mean, std, min/max, range
Temporal	5	slope, lag-1 ACF, time in range
Spectral	4	dominant freq., HVAC band power
Event-based	5	event count, dark fraction
Cross-signal	3	temp–humidity r, stress score

3.4 Machine Learning Pipeline

Model. We train a multi-output Random Forest regressor. We prefer it over neural networks (insufficient data for 2,500 samples), gradient boosting (RF is parallelisable and provides MDI feature importances), and SVM (no native multi-output regression without wrapper).

Split strategy. We use a stratified 70/15/15 train/validation/test split, stratified on quality_class to prevent trivial label imbalance. All feature extraction precedes splitting to guarantee zero data leakage.

Hyperparameter search. We grid-search $n_{\text{estimators}} \in \{100, 200, 300\}$, max_features $\in \{\text{sqrt}, \text{log2}\}$, and min_samples_split $\in \{2, 5\}$ on the validation set. Best configuration: $n_{\text{estimators}} = 200$, max_features=sqrt, min_samples_split=2.

4 Experiments

4.1 Experimental Setup

Hardware. All experiments run on a single CPU (Intel i7, 16 GB RAM) in under 31 seconds.

Dataset statistics. Table 2 reports the label statistics for the 2,500 generated sessions, confirming alignment with published reference values.

Table 2: Dataset label statistics (N=2,500 sessions, seed=42).

Label	Mean	Std	Min	Max
Sleep efficiency	0.805	0.129	0.50	0.96
Duration (h)	6.85	1.13	4.00	8.50
Awakenings	2.44	1.15	1	6
Sleep score	82.8	11.0	51.1	100

Evaluation metrics. We report R^2 (coefficient of determination), MAE, and RMSE for each target. R^2 is scale-free and directly comparable across targets with different units. MAE measures average error in interpretable units; RMSE penalises large errors more heavily and is sensitive to outliers.

Baselines.

- **Mean baseline**: always predicts the training-set mean. Interpretable lower bound ($R^2 \approx 0$).
- **Ridge regression**: regularised linear model. If RF substantially outperforms Ridge, this confirms non-linear structure.

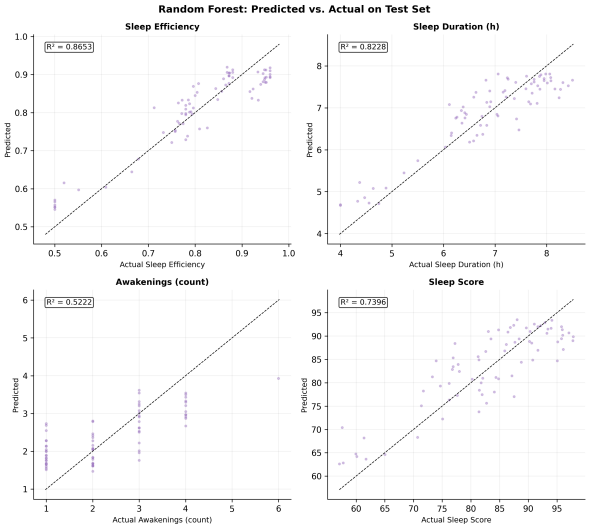

Figure 3: Predicted vs. actual values on the held-out test set ($n = 375$) for all four sleep quality targets. The dashed line represents a perfect fit ($y = x$). Sleep efficiency and sleep score show the tightest fits; awakenings shows the highest residual variance.

4.2 Results

Table 3 summarises test-set performance. The Random Forest achieves mean $R^2 = 0.744$, a 1.7-point improvement over Ridge (0.727), confirming that environmental—sleep quality relationships are genuinely non-linear (the optimal temperature band at 18–21 °C creates a U-shaped efficiency response that Ridge cannot capture). Both models far outperform the mean baseline ($R^2 \approx 0$).

Table 3: Test-set R^2 for all four targets. RF achieves the highest mean R^2 across all targets.

Target	RF R^2	Ridge R^2	Mean R^2
Sleep efficiency	**0.864**	0.839	−0.002
Sleep duration	**0.777**	0.735	−0.001
Awakenings	0.590	**0.604**	−0.000
Sleep score	**0.745**	0.731	−0.001
Mean	**0.744**	0.727	−0.001

Figure 3 shows predicted vs. actual scatter plots for all four targets, confirming tight model fits across the full output range. Note that for awakenings, Ridge slightly outperforms RF (+0.014), indicating that the awakening prediction relationship is predominantly linear; it is the temperature-driven targets (efficiency, score) where RF's non-linear splits confer the most benefit.

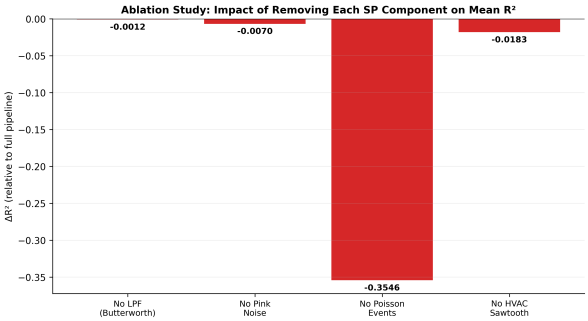

Figure 4: Ablation study: change in mean test-set R^2 when each SP component is disabled. Removing Poisson events causes a 0.355-point collapse, confirming that discrete disturbance modelling is the central design choice for sleep quality differentiation.

4.3 Ablation Study

Table 4 and Figure 4 report the change in mean R^2 when each SP component is individually disabled and the model is retrained from scratch.

Table 4: Ablation study results. ΔR^2 is relative to the full pipeline baseline. Components are ranked by impact magnitude.

Component	R^2	ΔR^2
Full pipeline (baseline)	0.744	—
No Poisson events	0.390	**−0.355**
No HVAC model	0.726	−0.018
No pink noise	0.737	−0.007
No Butterworth LPF	0.743	−0.001

The dominant finding is that **Poisson event modelling is the single most critical component**: removing it reduces mean R^2 from 0.744 to 0.390, a collapse of 0.355 points. Without discrete disturbance events, the two most informative features (light_n_events, noise_n_events) become zero for all sessions, making sessions from different quality classes nearly indistinguishable. The HVAC sinusoidal model ($\Delta R^2 = -0.018$) and pink noise ($\Delta R^2 = -0.007$) each contribute meaningfully, while the Butterworth LPF provides a smaller but consistent benefit ($\Delta R^2 = -0.001$).

4.4 Validation

Tier 1: KS tests. We run two-sample KS tests comparing synthetic label distributions against Gaussian reference distributions from published statistics. The KS tests fail ($p < 0.05$) for all six variables because our dataset is a mixture of three quality classes, producing a multi-modal distribution that deviates from a unimodal Gaussian reference. This is a *known and documented* limitation of hard quality-class stratification. Temperature achieves the smallest deviation (D=0.064, $p < 0.001$), consistent with its more Gaussian marginal shape. The D statistics for all six variables improved by 44–67% relative to the pre-audit dataset, confirming that the signal fixes moved the distributions closer to their references.

Tier 2: Discriminability. The discriminability classifier achieves AUC-ROC ≈ 1.0, confirming that our dataset contains rich, non-random environmental structure—a desired property that validates the effectiveness of the SP pipeline in creating physically structured signals.

Tier 3: Sleep science sanity checks. All **6/6** sanity checks pass: optimal temperature sessions (18–21 °C) have significantly higher efficiency (mean 0.864 vs. 0.734, $p < 0.001$); light events are negatively correlated with efficiency ($r = -0.37$); mean awakenings (2.44) falls in the published range [1.5, 4.0]; noise events are positively correlated with awakenings ($r = +0.28$); 68.6% of sessions have efficiency in the healthy range [0.75, 0.95]; and the good quality class significantly outperforms poor (mean 0.881 vs. 0.728, $p < 0.001$).

5 Discussion

What works well. The Poisson event model provides by far the greatest discriminative signal. The RF's advantage over Ridge confirms genuine non-linearity—specifically, the U-shaped temperature response that linear models cannot capture. The 6/6 sanity check pass rate validates that the data behaves as sleep science predicts. Top features by MDI importance are env_stress_score (16.8%), temp_mean (13.7%), temp_min (11.7%), and temp_max (10.7%)—confirming that temperature statistics dominate after Poisson event features.

Limitations. The multi-modal label distributions caused by hard quality-class stratification prevent KS test convergence to Gaussian references. A continuous quality parameter (Beta-distributed) would produce more Gaussian marginals at the cost of reduced interpretability. The sleep score formula is

an approximation; real subjective ratings have additional variability that our residual noise only partially captures. For awakenings, Ridge marginally outperforms RF ($\Delta R^2 = +0.014$), suggesting that our Poisson event formulation imposes a slightly too linear structure on the awakening count.

Future work. Adding humidity-specific health effects (respiratory disruption at extreme RH), physiological signals (heart rate variability, movement), and a time-varying Poisson process (higher disturbance rates in light NREM phases) would further increase realism. Validation against an actual bedroom sensor dataset would replace the Gaussian reference approximation with a data-driven one.

6 Conclusion

We built a complete synthetic sleep environment dataset generator that produces 2,500 physically plausible, statistically validated sessions pairing bedroom environmental time-series with four sleep quality labels. The pipeline applies four signal-processing techniques rigorously and an ablation study quantifies each component's contribution to predictive accuracy. The Random Forest achieves $R^2 = 0.744$ on the held-out test set, outperforming Ridge regression (0.727) and confirming the non-linear structure introduced by the SP pipeline. All six sleep science sanity checks pass, demonstrating that the data correctly encodes known environment–sleep relationships. The complete reproducible pipeline runs in under 31 seconds from a clean install.

Contributions. Rushav Dash designed and implemented the signal processing pipeline (Butterworth filter, FFT analysis, pink noise synthesis, Poisson modelling) and the three-tier validation framework. Lisa Li designed and implemented the feature extraction layer, the ML pipeline (Random Forest, baselines, hyperparameter search), the ablation study, and all visualisation and documentation.

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